

A New Ant Colony Algorithm Using the Heterarchical Concept Aimed at Optimization of Multim minima Continuous Functions

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Abstract. Ant colony algorithms are a class of metaheuristics which are inspired from the behaviour of real ants. The original idea consisted in simulating the trail communication, therefore these algorithms are considered as a form of adaptive memory programming. A new formalization is proposed for the design of ant colony algorithms, introducing the biological notions of heterarchy and communication channels.

1 Introduction

Having recently given raise to a new metaheuristic method, the ant colony metaphor proved to be a successful approach to solve “difficult” optimization problems. The first algorithm inspired from the ant colony functioning is the “ant system” (Colormi & al. 1991), which has been applied to many combinatorial problems. Until now, there are few adaptations of such algorithms to continuous optimization problems. The first algorithm designed for continuous function optimization was *CACO* (for *Continuous Ant Colony Optimization*) (Bilchev & al. 1995) which is close to the original metaphor, but unfortunately, the use of two different processes inside the *CACO* algorithm leads to a delicate setting of parameters.

All of these algorithms use a particular trait of real ants behaviour: the pheromonal trail laying. Indeed, ants colonies are often viewed as distributed systems capable of solving complex problems by the way of *stigmergy*, which is a form of indirect communication mediated by modifications of environment. But the trail-laying behaviour is also a part of the *recruitment* process, as the recruitment is defined by biologists as “a special form of assembly in which members of a society are directed to some point in space where work is required” (Hölldobler & Wilson 1990, p.642). According to this definition, another optimization method has been developed for continuous optimization: the *API* algorithm (Monmarché & al. 2000); but *API* makes a poor use of memory that generally characterizes ant colony systems (Taillard & al. 1998).

Our point of view is that ant colony metaphor can be defined as a model using trail communication or, more widely, as a recruitment process. According

to this idea, pheromonal trail laying may not be the only way to comprehend the ant colony metaphor for optimization problems (Bonabeau & al. 1997). Our recent research on modelling ants behaviour (Dréo 2001) has shown that the importance of inter-individuals communication may be underestimated, and that including it into an ant colony system may improve performances by accelerating the diffusion of information.

We were interested by these various ways of approaching the same idea and we thought that it would be useful to gather them in a single formalism. Indeed, there is another interesting way of approaching the social insects behaviour : the notion of *dense heterarchy*, developed in Sect. 2. An *heterarchical algorithm* takes advantages from a flow of information passing through a population of agents. These informations are exchanged using *communication channels* (see Sect. 3) and permit that a form of description of the objective function emerges from the system. In Sect. 4, we present an experimental result. Conclusion makes up the last section.

2 The Notion of Dense Heterarchy

2.1 A Biological Definition

This notion was firstly introduced by Wilson in 1988 to describe the information flow inside an ant colony (Wilson & Hölldobler 1988). In this idea, the two communication channels that we have evoked in the introduction are present, the stigmergic channel as well as the direct communication between individuals. One of the important issues here is that informations *flow* through the colony, each of the ants can communicate with any other (which makes the heterarchy being *dense*). This process constructs a kind of densely connected network which is not set up in a *hierarchical* but in a *heterarchical* manner (cf. Fig. 1).



Fig. 1. In a dense heterarchy, unlike hierarchy, any individual can communicate with any other one

Such a system has emergent properties, indeed if each agent operates with elementary rules and a limited accuracy, the whole population organized in heterarchy with mass communication can show an emergent pattern. These findings

are well known in the study of self-organization (Camazine & al. 2000). To summarize, the notion of dense heterarchy describes the way the ant colony handles the informations that it receives from the environment. Each ant can communicate with any other at any time and the informations flow through the colony.

2.2 Heterarchical Algorithms

We propose here a simple formalization in order to apply the notion of dense heterarchy to an optimization problem. The main concept for implementing such an *heterarchical algorithm* is the idea of *communication channel* (cf. Fig. 2). These channels transmit an information, here the localization of a food source, and have some properties, like stigmergy and memory, which can be combined in the same channel, so that a large variety of different channels can be built.

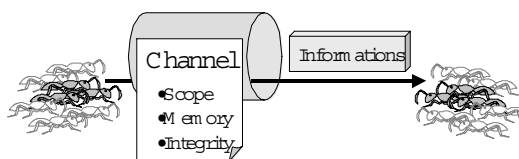


Fig. 2. The informations pass from a part of the population to another one through a communication channel that has some specific properties

For example, some basic properties of the channels are listed below:

1. *Scope*: the way the information goes through the population. A sub-group of the population (from one to n agents) can exchange informations with another group of agents.
2. *Memory*: the way the information persists in the system. The information can be stored during some period of time or be transitory.
3. *Integrity*: the way the information is evolving in the system. The information can be modified, by one or more agents, by an external process, or not.

3 The Communication Channels

3.1 The Trail Channel

The first version of our algorithm is trying to be as close as possible to a continuous version of the original *ACO* (*Ant Colony Optimization*) algorithm (Colormi & al. 1991) which was designed for combinatorial problems. Due to this assumption, it has somewhat the same design as the local search part of the *CACO* algorithm (Bilchev & al. 1995), which was also inspired by the first Ant System.

This implementation uses only one communication channel which is inspired from the trail laying behaviour of ants. Here, each ant can deposit a certain

amount of *pheromone* as a spot on the search space, proportionally to the amelioration of the objective function she finds on her way. These pheromonal spots can be perceived by all the members of the population, and diffuse into the environment. The ants are attracted by each spot according to its distance and to the amount of pheromone it contains. Each artificial ant has a “range” parameter that is normally distributed over the population. Each ant draws a random distance and then jumps of this length in the direction of her weighted gravity center, some noise modifying the final position.

To summarize this behaviour under the vision of the heterarchical concept, this stigmergic communication channel shows the following properties, underlined in Sect. 2.2:

1. *Scope*: When one ant lays a pheromonal spot, all the ants can subsequently perceive it.
2. *Memory*: The information persists in the system during a certain period of time, independently of the agents.
3. *Integrity*: The information is modified by time, to reproduce the pheromone evaporation.

There is some similarity between this algorithm and the “path-relinking” algorithm introduced by Glover (Glover & Laguna 1997), as the ants are moving through a set of informative points. We can also notice that the behavior of the algorithm reminds the *Particle Swarm Optimization* (PSO Kennedy & al. 1995) as the agents tend to gather in the same place, the whole population evolving as a swarm.

3.2 Using the Direct Inter-individual Communication

As we have said in Sect. 1, some biological works have led us to take an interest out of the common vision of the “trail” ant colony optimization. Indeed, we have implemented another communication channel which possesses the properties of the direct inter-individual interactions that can be observed in societies of some social insects as ants.

In concrete terms, each ant can send “messages” to another one. An ant receiving a message stores it in a stack with other incoming messages. In a second time, a message is read randomly in the stack. Here, the information sent is the position of the sender – an artificial ant – and the value of the objective function. The receiver compares the sender’s value with its own value and decides if it moves near the sender’s position. But if the receiver’s value is better than the sender’s one, then the receiver sends a message to another ant randomly chosen, and suppresses the read message. One can notice that the system needs to be “activated”, so that an important parameter is the number of messages initially set.

This communication channel shows the following properties:

1. *Scope*: When an ant sends a message, only one ant can perceive it.
2. *Memory*: The information persists in the system during a certain period of time, under the form of ants memory.
3. *Integrity*: The informations stored are static.

4 Experimental Result

To illustrate the behaviour of an heterarchical algorithm, we have applied it on the B_2 function, which has two dimensions and some local minima. The two channels play complementary parts. Indeed, the direct channel leads to a form of intensification, because it gives more importance to the best points without taking into account the previously encountered regions. On the contrary, the trail channel—with its memory property—permits to perform a kind of diversification, by taking into account the previously evaluated points. Globally, the ants gather at the global optimum. However some of them are kept during some iterations near local optima by some phomonal spots still persisting within the search space, afterwards evaporation and direct communications prevent them from being trapped. The way the trail channel permits a form of diversification is pointed out with more difficulties. The ants are moving around a gravity center that is the global optimum, but continue to explore the search space, as they are attracted by local minima.

One interesting issue is the way the two channels works in *synergy*. Indeed, on the simple B_2 function, the direct channel seems to be more appropriate as it allows the algorithm to converge more rapidly (Fig. 3). But if we take a look to the variation of the standard deviation for the algorithm using the two communication channels (Fig. 3), we notice that there is a kind of *periodicity*. This means that the population tends to gather near a value at one time and then tends to disperse. In other words, there is an alternation of short intensification phases (low deviation) and diversification phases (high deviation). This behaviour of the algorithm seems to be an emergent pattern, which cannot be observed when only one of the two channels is used. Thus the algorithm regulates itself the way the two channels are working, until a stable state is found.

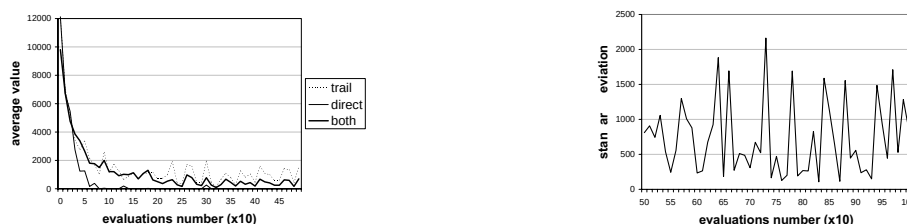


Fig. 3. Average and standard deviation of the objective function values in the population at different steps

5 Conclusion

We argue that the heterarchical concept can be interesting to design new ant colony algorithms, in particular aimed at the optimization of continuous multim minima functions. Such a biological concept was not exploited until now for

the design of optimization algorithms, using only stigmergic processes. We propose to extend the ant colony metaphor to take into account several communication processes. According to our first tests it seems that an heterarchical algorithm implementing two complementary communication channels shows interesting emergent properties, like a self-management of the relative influence of the two channels.

Regarding the future, one important issue consists in improving the automatic tuning of parameters, before testing the algorithm through a large set of analytical test functions, and comparing its efficiency to that of competing metaheuristics.

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